Jesse S. Jing

Washington D.C. Area | Atlanta, GA | (301) 602-7789 | jjing43@gatech.edu | U.S. Citizen

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
B.S. in Computer Engineering Expected Graduation: May 2029

Coursework (In Progress): Intro to Object-Oriented Programming (Java), Discrete Mathematics, Linear Algebra
Coursework (Spring 2027): Algorithms & Data Structures, Multivariable Calculus, Fundamentals of Digital System Design

Montgomery Blair High School | Silver Spring, MD *August 2022 – May 2026*
Science, Mathematics, and Computer Science Magnet Program, GPA 3.97/4.0 UW

Research Experience

Beller Lab | Baltimore, Maryland *June 2025 – August 2025*
Paid Research Intern / JHU Bloomberg Center for Physics & Astronomy

- Developed a Python data-analysis pipeline using KD-trees and graph-based clustering to quantify emergent patterns in active matter simulations of cytoskeletal filaments
- Analyzed spatial correlations and cluster dynamics across model parameters, identifying a novel active state in which filaments form but destabilize nematic lanes
- Presented findings at the JHU REAL Undergraduate Showcase and the APS Southeastern Section Annual Meeting

Porto Lab | College Park, Maryland *June 2025 – August 2025*
Research Intern / UMD Department of Physics

- Engineered the first computational prototype for a quantum optics simulation framework in Python, modeling complex wave propagation for proposed optical setups
- Identified performance bottlenecks and tested several optimization approaches; results were incorporated into a graduate researcher's PhD candidacy proposal

Sternbach Lab | College Park, Maryland *November 2023 – September 2024*
Research Intern / UMD Department of Physics

- Built a database of nonlinear optical properties of organic crystals to support quantum materials research
- Used LabVIEW to investigate near-field/far-field imaging in atomic force microscopy (AFM) experiments

Skills

Programming: Java, Python | **Hardware Description:** SystemVerilog | **Tools:** LabVIEW
Languages: English (fluent), Chinese (conversational), Spanish (conversational)

Leadership & Activities

Montgomery Blair High School | Physics Team Captain *August 2024 – May 2025*

- Led weekly physics lectures and demos for 40+ members; grew club membership by 40%
- Created and maintained a physics tutoring server for 140+ active members

Montgomery Blair High School | MBMT Task Force Lead *August 2024 – May 2026*

- Led organization of the county's largest math competition for 400+ middle school students
- Wrote competition problems and managed planning; secured $16,000+ from 10+ sponsors through outreach

Honors & Awards

2026 - National Merit $2,500 Scholarship Recipient
2025 - USACO Silver Division
2025 - ACSL Senior Division Finalist
2024 - High School Mathematical Contest in Modeling (HiMCM) Honorable Mention